

Adjoint Boundary Control for a Bi-harmonic type of System-Calculus of Variation

Tags: #CFD #Variation #Fluid-Mechanics

Introduction

In this note, we are going to introduce how to derive adjoint equations from Lagrangian functional and calculus of variation for Bi-harmonic type of system.

Notion of adjoint equations

Problem set: An abstract optimization problem (Marx D. Gunzburger, page 12)

Let ϕ denote the state variable, g the control variables or design parameters, $J(\phi, g)$ the cost or objective functional, and $F(\phi, g)$. The optimization problem is: **Find controls g and states ϕ such that $J(\phi, g)$ is minimized subject to $F(\phi, g) = 0$**

Remark

For the following steps, we assume that the functionals and variables are all sufficiently smooth.

Let us introduce the adjoint variable (Lagrange multiplier) and the Lagrangian functional

$$\mathcal{L}(\phi, g, \xi) = J(\phi, g) - \xi^* F(\phi, g)$$

Then the constrained optimization is converted to the unconstrained optimization problem: **Find controls g , states ϕ , and co-states ξ such that $\mathcal{L}(\phi, g, \xi)$ is rendered stationary (i.e $\delta \mathcal{L} = 0$).**

Remark

Note that each argument of $\mathcal{L}(\phi, g, \xi)$ is considered to be independent of the others; this was not true for the original optimization problem involving $J(\phi, g)$ since there the arguments ϕ and g were constrained to satisfy $F(\phi, g) = 0$ and thus could not be chosen independently.

The adjoint equations are obtained from the first variations of the Lagrangian function :

$$\lim_{\varepsilon \rightarrow 0} \left(\frac{\mathcal{L}(\phi + \varepsilon \tilde{\phi}, g, \xi) - \mathcal{L}(\phi, g, \xi)}{\varepsilon} \right) = \lim_{\varepsilon \rightarrow 0} \left(\frac{J(\phi + \varepsilon \tilde{\phi}, g) - J(\phi, g)}{\varepsilon} - \frac{\xi^*(F(\phi + \varepsilon \tilde{\phi}, g) - F(\phi, g))}{\varepsilon} \right) = 0$$

By the Taylor series

$$J(\phi + \varepsilon \tilde{\phi}, g) = J(\phi, g) + \varepsilon \left(\frac{\partial J}{\partial \phi} \Big|_{(\phi, g)} \right) \tilde{\phi} + O(\varepsilon^2)$$

$$F(\phi + \varepsilon \tilde{\phi}, g) = F(\phi, g) + \varepsilon \left(\frac{\partial F}{\partial \phi} \Big|_{(\phi, g)} \right) \tilde{\phi} + O(\varepsilon^2)$$

Thus

$$\lim_{\varepsilon \rightarrow 0} \left(\left(\frac{\partial J}{\partial \phi} \Big|_{(\phi, g)} \right) \tilde{\phi} - \xi^* \left(\frac{\partial F}{\partial \phi} \Big|_{(\phi, g)} \right) \tilde{\phi} + O(\varepsilon) \right) = 0$$

and

$$\left(\frac{\partial J}{\partial \phi} \Big|_{(\phi, g)} \right) \tilde{\phi} - \xi^* \left(\frac{\partial F}{\partial \phi} \Big|_{(\phi, g)} \right) \tilde{\phi} = 0 \iff \tilde{\phi}^* \left(\left(\frac{\partial J}{\partial \phi} \Big|_{(\phi, g)} \right)^* - \left(\frac{\partial F}{\partial \phi} \Big|_{(\phi, g)} \right)^* \xi \right) = 0$$

The adjoint equation is

$$\left(\frac{\partial J}{\partial \phi} \Big|_{(\phi, g)} \right)^* - \left(\frac{\partial F}{\partial \phi} \Big|_{(\phi, g)} \right)^* \xi = 0$$

With the similar idea, we can obtain the optimality condition:

$$\lim_{\varepsilon \rightarrow 0} \left(\frac{\mathcal{L}(\phi, g + \varepsilon \tilde{g}, \xi) - \mathcal{L}(\phi, g, \xi)}{\varepsilon} \right) = \lim_{\varepsilon \rightarrow 0} \left(\frac{J(\phi, g + \varepsilon \tilde{g}) - J(\phi, g)}{\varepsilon} - \frac{\xi^*(F(\phi, g + \varepsilon \tilde{g}) - F(\phi, g))}{\varepsilon} \right) = 0$$

Applying the Taylor series, we have

$$\lim_{\varepsilon \rightarrow 0} \left(\left(\frac{\partial J}{\partial g} \Big|_{(\phi, g)} \right) \tilde{g} - \xi^* \left(\frac{\partial F}{\partial g} \Big|_{(\phi, g)} \right) \tilde{g} + O(\varepsilon) \right) = 0$$

Thus the optimality condition is

$$\left(\frac{\partial J}{\partial g} \Big|_{(\phi, g)} \right)^* - \left(\frac{\partial F}{\partial g} \Big|_{(\phi, g)} \right)^* \xi = 0$$

To update the control variable or control variables in the iterations, we need to resolve the *gradient of functional* and use it with the gradient descent method. The procedure of determining the gradient of functional is shown in the following example.

Example: A control problem with infinite-dimensional controls (Marx D. Gunzburger, page 33)

Let Ω be an open domain in $\mathbb{R}^n$, $n = 2, 3$, having boundary $\partial\Omega$. Let ϕ denote the scalar-valued state variable and f and g denote the infinite-dimensional scalar-valued control functions. The objective functional is given by

$$J(\phi, f, g) = \frac{1}{2} \int_{\Omega} (\phi - \Phi)^2 d\Omega + \frac{\sigma_1}{2} \int_{\Omega} f^2 d\Omega + \frac{\sigma_2}{2} \int_{\partial\Omega} g^2 d\Gamma$$

and the constraints are

$$\begin{aligned} -\nabla^2 \phi + \phi^3 &= f \quad \text{in } \Omega \\ \frac{\partial \phi}{\partial n} &= g \quad \text{on } \partial\Omega \end{aligned}$$

We want to find ϕ , f and g such that the functional J is minimized subject to the requirements that ϕ , f and g satisfy the constraints.

We introduce a Lagrangian functional such that

$$\mathcal{L}(\phi, f, g, \xi, \eta) = J(\phi, f, g) - \int_{\Omega} \xi(-\nabla^2 \phi + \phi^3 - f) d\Omega - \int_{\partial\Omega} \eta(\frac{\partial \phi}{\partial n} - g) d\Gamma$$

Applying the first variation of the Lagrangian functional, we obtain the state equations

$$\begin{aligned} -\nabla^2 \phi + \phi^3 &= f \quad \text{in } \Omega \\ \frac{\partial \phi}{\partial n} &= g \quad \text{on } \partial\Omega \end{aligned}$$

and the adjoint equations

$$\begin{cases} -\nabla^2 \xi + 3\phi^2 \xi = \phi - \Phi & \text{in } \Omega \\ \frac{\partial \xi}{\partial n} = 0 & \text{in } \partial\Omega \end{cases}$$

We can acquire the gradients functional for f and g with the adjoint equations, but we need to determine the sensitivity equations to do that.

The sensitivity equation for f

The weak variations $\bar{f}$ and $\bar{\phi}$ are defined as

$$\begin{aligned} \bar{f} &= f + \varepsilon \tilde{f} \\ \bar{\phi}_f &= \phi + \varepsilon \phi_f \end{aligned}$$

where ϕ_f represents the perturbation induced by the variation of f . Substituting the expressions

of $\bar{f}$ and $\bar{\phi}_f$ into the state equation, we have

$$\begin{aligned} -\nabla^2(\phi + \varepsilon\phi_f) + (\phi + \varepsilon\phi_f)^3 &= f + \varepsilon\tilde{f} \quad \text{in } \Omega \\ \frac{\partial(\phi + \varepsilon\phi_f)}{\partial n} &= g \quad \text{on } \partial\Omega \end{aligned}$$

Developing those equations, they are reduced to

$$\begin{aligned} -\nabla^2\phi_f + 3\phi_f\phi^2 + 3\varepsilon\phi_f^2\phi + \varepsilon^2\phi_f^3 &= \tilde{f} \quad \text{in } \Omega \\ \frac{\partial\phi}{\partial n} + \varepsilon\frac{\partial\phi_f}{\partial n} &= g \quad \text{on } \partial\Omega \end{aligned}$$

Thus

$$\frac{\partial\phi_f}{\partial n} = 0 \quad \text{on } \partial\Omega$$

and by setting ε tends to zero, we have

$$-\nabla^2\phi_f + 3\phi^2\phi_f = \tilde{f} \quad \text{in } \Omega$$

which is the sensitivity equation for f .

The sensitivity equation for g

The weak variations g and ϕ are defined as

$$\begin{aligned} \bar{g} &= g + \varepsilon\tilde{g} \\ \bar{\phi}_g &= \phi + \varepsilon\phi_g \end{aligned}$$

where ϕ_g represents the perturbation induced by the variation of g . Substituting the expressions of $\bar{g}$ and $\bar{\phi}_g$ into the state equation, we have

$$\begin{aligned} -\nabla^2(\phi + \varepsilon\phi_g) + (\phi + \varepsilon\phi_g)^3 &= f \quad \text{in } \Omega \\ \frac{\partial(\phi + \varepsilon\phi_g)}{\partial n} &= g + \varepsilon\tilde{g} \quad \text{on } \partial\Omega \end{aligned}$$

Developing those equations, they are reduced to

$$\begin{aligned} -\nabla^2\phi_g + 3\phi_g\phi^2 + 3\varepsilon\phi_g^2\phi + \varepsilon^2\phi_g^3 &= 0 \quad \text{in } \Omega \\ \frac{\partial\phi}{\partial n} + \varepsilon\frac{\partial\phi_g}{\partial n} &= g + \varepsilon\tilde{g} \quad \text{on } \partial\Omega \end{aligned}$$

Thus

$$\frac{\partial\phi_g}{\partial n} = \tilde{g} \quad \text{on } \partial\Omega$$

and by setting ε tends to zero, we have

$$-\nabla^2 \phi_g + 3\phi^2 \phi_g = 0 \quad \text{in } \Omega$$

which is the sensitivity equation for g .

Since the first variation with respect to the variations of f and g can be defined as

$$\delta J = (\nabla_f J \cdot \tilde{f} + \nabla_g J \cdot \tilde{g})\varepsilon = \left(\int_{\Omega} \nabla_f J \tilde{f} d\Omega + \int_{\partial\Omega} \nabla_g J \tilde{g} d\Gamma \right) \varepsilon$$

where $\nabla_f J$ and $\nabla_g J$ are the gradients of functional that we need. There are two approaches to determine them. The first approach is by calculating the first variation separately, such that

$$\begin{aligned} \nabla_f J \cdot \tilde{f} &= \lim_{\varepsilon \rightarrow 0} \left(\frac{J(\phi + \varepsilon \phi_f, f + \varepsilon \tilde{f}, g) - J(\phi, f, g)}{\varepsilon} \right) \\ \nabla_g J \cdot \tilde{g} &= \lim_{\varepsilon \rightarrow 0} \left(\frac{J(\phi + \varepsilon \phi_g, f, g + \varepsilon \tilde{g}) - J(\phi, f, g)}{\varepsilon} \right) \end{aligned}$$

which is equivalent to

$$\begin{aligned} J(\varepsilon) &= J(\phi + \varepsilon(\phi_f + \phi_g), f + \varepsilon \tilde{f}, g + \varepsilon \tilde{g}) \\ &= \frac{1}{2} \int_{\Omega} (\phi + \varepsilon(\phi_f + \phi_g) - \Phi)^2 d\Omega + \frac{\sigma_1}{2} \int_{\Omega} (f + \varepsilon \tilde{f})^2 d\Omega + \frac{\sigma_2}{2} \int_{\partial\Omega} (g + \varepsilon \tilde{g})^2 d\Gamma \end{aligned}$$

and computing the derivative with respect to ε such that

$$\begin{aligned} \left. \frac{dJ(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} &= \int_{\Omega} (\phi_f + \phi_g)(\phi - \Phi) d\Omega + \sigma_1 \int_{\Omega} f \tilde{f} d\Omega + \sigma_2 \int_{\partial\Omega} g \tilde{g} d\Gamma \\ &= \int_{\Omega} \phi_f(\phi - \Phi) d\Omega + \sigma_1 \int_{\Omega} f \tilde{f} d\Omega + \int_{\Omega} \phi_g(\phi - \Phi) d\Omega + \sigma_2 \int_{\partial\Omega} g \tilde{g} d\Gamma \\ &= \nabla_f J \cdot \tilde{f} + \nabla_g J \cdot \tilde{g} \end{aligned}$$

In order to acquire the gradients of functional with the adjoint equations, we need to substitute the state equation into the integrals and apply the (second) Green's identity.

The gradient of functional with respect to f

By the previous expression, we have

$$\nabla_f J \cdot \tilde{f} = \int_{\Omega} \phi_f(\phi - \Phi) d\Omega + \sigma_1 \int_{\Omega} f \tilde{f} d\Omega$$

According to the state equation ,we have

$$\int_{\Omega} \phi_f(\phi - \Phi) d\Omega = \int_{\Omega} \phi_f(-\nabla^2 \xi + 3\phi^2 \xi) d\Omega$$

where

$$\begin{aligned}\frac{\partial \xi}{\partial n} &= 0 \quad \text{on } \partial\Omega \\ \frac{\partial \phi_f}{\partial n} &= 0 \quad \text{on } \partial\Omega\end{aligned}$$

By Green's identity, we have

$$\int_{\Omega} (\xi \nabla^2 \phi_f - \phi_f \nabla^2 \xi) d\Omega = \int_{\partial\Omega} \left(\xi \frac{\partial \phi_f}{\partial n} - \phi_f \frac{\partial \xi}{\partial n} \right) d\Gamma$$

Thus

$$-\int_{\Omega} \phi_f \nabla^2 \xi d\Omega = -\int_{\Omega} \xi \nabla^2 \phi_f d\Omega$$

and the sensitivity equation renders us

$$-\xi \nabla^2 \phi_f = (\tilde{f} - 3\phi^2 \phi_f) \xi$$

Therefore

$$\begin{aligned}\nabla_f J \cdot \tilde{f} &= \int_{\Omega} \phi_f (\phi - \Phi) d\Omega + \sigma_1 \int_{\Omega} f \tilde{f} d\Omega \\ &= \int_{\Omega} ((\tilde{f} - 3\phi^2 \phi_f) \xi + 3\phi^2 \phi_f \xi) d\Omega + \sigma_1 \int_{\Omega} f \tilde{f} d\Omega \\ &= \int_{\Omega} (\xi + \sigma_1 f) \tilde{f} d\Omega\end{aligned}$$

The gradient of functional with respect to g

By the previous expression, we obtain

$$\nabla_g J \cdot \tilde{g} = \int_{\Omega} \phi_g (\phi - \Phi) d\Omega + \sigma_2 \int_{\partial\Omega} g \tilde{g} d\Gamma$$

According to the state equation, we have

$$\int_{\Omega} \phi_g (\phi - \Phi) d\Omega = \int_{\Omega} \phi_g (-\nabla^2 \xi + 3\phi^2 \xi) d\Omega$$

where

$$\begin{aligned}\frac{\partial \xi}{\partial n} &= 0 \quad \text{on } \partial\Omega \\ \frac{\partial \phi_g}{\partial n} &= \tilde{g} \quad \text{on } \partial\Omega\end{aligned}$$

By Green's identity, we obtain

$$\int_{\Omega} (\nabla^2 \phi_g \xi - \phi_g \nabla^2 \xi) d\Omega = \int_{\partial\Omega} \left(\xi \frac{\partial \phi_g}{\partial n} - \phi_g \frac{\partial \xi}{\partial n} \right) d\Gamma = \int_{\partial\Omega} \xi \tilde{g} d\Gamma$$

Thus

$$-\int_{\Omega} \phi_g \nabla^2 \xi \, d\Omega = -\int_{\Omega} \xi \nabla^2 \phi_g \, d\Omega + \int_{\partial\Omega} \xi \tilde{g} \, d\Gamma$$

Therefore

$$\begin{aligned} \nabla_g J \cdot \tilde{g} &= \int_{\Omega} \phi_g (\phi - \Phi) \, d\Omega + \sigma_2 \int_{\partial\Omega} g \tilde{g} \, d\Gamma \\ &= \int_{\Omega} (3\phi^2 \phi_g \xi - 3\phi^2 \phi_g \xi) \, d\Omega + \int_{\partial\Omega} (\sigma_2 g \tilde{g} + \xi \tilde{g}) \, d\Gamma \\ &= \int_{\partial\Omega} (\xi + \sigma_2 g) \tilde{g} \, d\Gamma \end{aligned}$$

Important Conclusions

- The adjoint system is independent of the particular choice of changes $\tilde{f}$ and $\tilde{g}$ in the controls.
- As a result, no matter how many changes $\tilde{f}$ and $\tilde{g}$ in the controls one chooses, one can use gradients of functional to determine the corresponding change in the functional with **only a single solution of the adjoint system**.

Application to the two-dimensional steady state Stokes flow

Problem Setting

The stream function-vorticity formulation of a steady state Stokes flow is written as

$$\begin{cases} \nabla^2 \psi = \omega \\ \nabla^2 \omega = 0 \end{cases}$$

with the Dirichlet and Neumann boundary conditions of ψ

$$\begin{aligned} \psi|_{\partial\Omega} &= f(x) \\ \frac{\partial \psi}{\partial n} \Big|_{\partial\Omega} &= \mathbf{u}_{\Gamma} \cdot \mathbf{t} \end{aligned}$$

where

$$\Gamma = \partial\Omega$$

is the boundary of the domain Ω and $\mathbf{t}$ is the tangent vector along the boundary curve. The objective functional of this problem is given by

$$J(\tilde{\psi}, u) = \frac{1}{2} \int_{\partial\Omega} \left(\frac{\partial \tilde{\psi}}{\partial n} - \mathbf{u}_{\Gamma} \cdot \mathbf{t} \right)^2 ds + \frac{\alpha}{2} \int_{\partial\Omega} \eta^2 ds$$

subject to the constraints

$$\begin{aligned}\nabla^2 \tilde{\psi} - \omega &= 0 \quad \text{in } \Omega \\ \nabla^2 \omega &= 0 \quad \text{in } \Omega\end{aligned}$$

where η is the control variable

$$\eta = \omega|_{\partial\Omega}$$

We introduce a Lagrangian functional such that

$$\mathcal{L}(\tilde{\psi}, \omega, \eta, p, q) = \frac{1}{2} \int_{\partial\Omega} \left(\frac{\partial \tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma \cdot \mathbf{t} \right)^2 ds + \frac{\alpha}{2} \int_{\partial\Omega} \eta^2 ds + \int_{\Omega} p(\nabla^2 \tilde{\psi} - \omega) d\Omega + \int_{\Omega} q(\nabla^2 \omega) d\Omega$$

and we define weak variations such that

$$\begin{aligned}\tilde{\psi}^* &= \tilde{\psi} + \varepsilon \delta \tilde{\psi} \\ \omega^* &= \omega + \varepsilon \delta \omega \\ \eta^* &= \eta + \varepsilon \delta \eta\end{aligned}$$

where

$$\begin{aligned}\delta \tilde{\psi}|_{\partial\Omega} &= 0 \\ \delta \omega|_{\partial\Omega} &= \delta \eta\end{aligned}$$

Applying the calculus of the first variation to the Lagrangian function, we determine the adjoint equations such that

$$\begin{aligned}\nabla^2 p &= 0 \quad \forall x \in \Omega \\ \nabla^2 q &= p \quad \forall x \in \Omega \\ p &= -\left(\frac{\partial \tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma \cdot \mathbf{t} \right) \quad \forall x \in \partial\Omega \\ q &= 0 \quad \forall x \in \partial\Omega\end{aligned}$$

Now we have to deduce the sensitivity equations before determining the gradient of the objective functional. Applying the weak variations to the state equations, it yields

$$\begin{aligned}\nabla^2(\tilde{\psi} + \varepsilon \delta \tilde{\psi}) &= \omega + \varepsilon \delta \omega \\ \nabla^2(\omega + \varepsilon \delta \omega) &= 0\end{aligned}$$

This implies

$$\begin{aligned}\nabla^2 \delta \tilde{\psi} &= \delta \omega \\ \nabla^2 \delta \omega &= 0\end{aligned}$$

which are the sensitivity equations.

Since $\delta\tilde{\psi}$ is induced by the δu , thus we should have

$$\begin{aligned}\left.\frac{dJ(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} &= \lim_{\varepsilon \rightarrow 0} \frac{J(\tilde{\psi} + \varepsilon \delta\tilde{\psi}, \eta + \varepsilon \delta\eta) - J(\tilde{\psi}, \eta)}{\varepsilon} \\ &= \nabla_{\eta} J \cdot \delta\eta \\ &= \int_{\partial\Omega} \nabla_{\eta} J \delta\eta \, ds\end{aligned}$$

where

$$\left.\frac{dJ(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \int_{\partial\Omega} \left(\frac{\partial\tilde{\psi}}{\partial n} - \mathbf{u}_{\Gamma} \cdot \mathbf{t}\right) \left(\frac{\partial\delta\tilde{\psi}}{\partial n}\right) ds + \int_{\partial\Omega} \alpha\eta \delta\eta \, ds$$

Applying the adjoint equation, it provides us

$$\int_{\partial\Omega} \left(\frac{\partial\tilde{\psi}}{\partial n} - \mathbf{u}_{\Gamma} \cdot \mathbf{t}\right) \left(\frac{\partial\delta\tilde{\psi}}{\partial n}\right) ds = \int_{\partial\Omega} -p \frac{\partial\delta\tilde{\psi}}{\partial n} \, ds$$

By the Green's identity, we have

$$\int_{\Omega} (\nabla^2 p \delta\tilde{\psi} - p \nabla^2 \delta\tilde{\psi}) \, d\Omega = \int_{\partial\Omega} \left(\frac{\partial p}{\partial n} \delta\tilde{\psi} - p \frac{\partial\delta\tilde{\psi}}{\partial n}\right) ds$$

which implies with the sensitivity equation

$$\int_{\partial\Omega} -p \frac{\partial\delta\tilde{\psi}}{\partial n} \, ds = \int_{\Omega} -p \nabla^2 \delta\tilde{\psi} \, d\Omega = \int_{\Omega} -p \delta\omega \, d\Omega$$

Substitute the adjoint equation into this relation, we have

$$\int_{\Omega} -p \delta\omega \, d\Omega = \int_{\Omega} -\nabla^2 q \delta\omega \, d\Omega$$

Using the Green's identity once again, it renders us

$$\int_{\Omega} (q \nabla^2 \delta\omega - \delta\omega \nabla^2 q) \, d\Omega = \int_{\partial\Omega} \left(q \frac{\partial\delta\omega}{\partial n} - \frac{\partial q}{\partial n} \delta\omega\right) ds$$

Therefore, we obtain

$$\int_{\Omega} (\nabla^2 p \delta\tilde{\psi} - p \nabla^2 \delta\tilde{\psi}) \, d\Omega = \int_{\Omega} -\delta\omega \nabla^2 q \, d\Omega = \int_{\partial\Omega} -\frac{\partial q}{\partial n} \delta\omega \, ds$$

and

$$\nabla_{\eta} J \cdot \delta\eta = \int_{\partial\Omega} \left(-\frac{\partial q}{\partial n} + \alpha\eta\right) \delta\eta \, ds$$

which is the gradient of the objective functional.

Application to the two-dimensional steady state Navier-Stokes equation

Problem Setting

The stream function-vorticity formulation of a steady state Navier-Stokes equation is written as

$$\begin{cases} \nabla^2 \psi = \omega \\ \nabla^2 \omega = Re(u \cdot \nabla) \omega \end{cases}$$

with the Dirichlet and Neumann boundary conditions of ψ

$$\begin{aligned} \psi|_{\partial\Omega} &= f(x) \\ \frac{\partial \psi}{\partial n} \Big|_{\partial\Omega} &= \mathbf{u}_\Gamma \cdot \mathbf{t} \end{aligned}$$

where

$$\Gamma = \partial\Omega$$

is the boundary of the domain Ω and $\mathbf{t}$ is the tangent vector along the boundary curve. The objective functional of this problem is given by

$$J(\tilde{\psi}, u) = \frac{1}{2} \int_{\partial\Omega} \left(\frac{\partial \tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma \cdot \mathbf{t} \right)^2 ds + \frac{\alpha}{2} \int_{\partial\Omega} \eta^2 ds$$

subject to the constraints

$$\begin{aligned} \nabla^2 \tilde{\psi} - \omega &= 0 \quad \text{in } \Omega \\ \nabla^2 \omega - Re(u \cdot \nabla) \omega &= 0 \quad \text{in } \Omega \\ u &= -\nabla \times \psi \mathbf{e}_z \quad \text{in } \Omega \end{aligned}$$

where η is the control variable

$$\eta = \omega|_{\partial\Omega}$$

We define a Lagrangian functional such that

$$\begin{aligned} \mathcal{L}(\tilde{\psi}, \omega, u, \eta, \beta, \gamma, \lambda) &= J(\tilde{\psi}, \eta) + R(\tilde{\psi}, \omega, \beta) + S(\omega, u, \gamma) + T(\tilde{\psi}, u, \lambda) \\ &= \frac{1}{2} \int_{\partial\Omega} \left(\frac{\partial \tilde{\psi}}{\partial n} - a \right)^2 ds + \frac{\alpha}{2} \int_{\partial\Omega} \eta^2 ds \\ &\quad + \int_{\Omega} \beta (\nabla^2 \tilde{\psi} - \omega) d\Omega + \int_{\Omega} \gamma (\nabla^2 \omega - Re(u \cdot \nabla) \omega) d\Omega \\ &\quad + \int_{\Omega} \lambda \cdot (\nabla \times \tilde{\psi} \mathbf{e}_z + u) d\Omega \end{aligned}$$

The weak variations of system states and control variable are defined as

$$\begin{aligned}\tilde{\psi}^* &= \tilde{\psi} + \varepsilon \delta \tilde{\psi} \\ \omega^* &= \omega + \varepsilon \delta \omega \\ \eta^* &= \eta + \varepsilon \delta \eta \\ u^* &= u + \varepsilon \delta u\end{aligned}$$

The adjoint equations are derived from the calculus of variation such that

$$\begin{aligned}\frac{\partial \tilde{\psi}}{\partial n} - a + \beta &= 0 \quad \text{on } \partial\Omega \\ \nabla^2 \beta &= -Re(\nabla \omega \times \nabla \gamma) \cdot \mathbf{e}_z \quad \text{in } \Omega \\ \nabla^2 \gamma + Re(u \cdot \nabla) \gamma &= \beta \quad \text{in } \Omega \\ \gamma &= 0 \quad \text{on } \partial\Omega\end{aligned}$$

where $a = \mathbf{u}_\Gamma \cdot \mathbf{t}$ is the tangential component of boundary velocity.

The gradient of the objective functional is

$$\nabla_\eta J = \alpha \eta - \frac{\partial \gamma}{\partial n}$$

To prove that it is indeed the objective functional, let us consider the derivative of first variation of the objective functional and substitute the adjoint equations into it, and we will obtain

$$\begin{aligned}\left. \frac{dJ}{d\varepsilon} \right|_{\varepsilon=0} &= \int_{\partial\Omega} \left(\frac{\partial \tilde{\psi}}{\partial n} - a \right) \frac{\partial \delta \tilde{\psi}}{\partial n} ds + \int_{\partial\Omega} \alpha \eta \delta \eta ds \\ &= \int_{\partial\Omega} -\beta \frac{\partial \delta \tilde{\psi}}{\partial n} ds + \int_{\partial\Omega} \alpha \eta \delta \eta ds\end{aligned}$$

where

$$\begin{aligned}\int_{\partial\Omega} -\beta \frac{\partial \delta \tilde{\psi}}{\partial n} ds &= \int_{\Omega} (\nabla^2 \beta \delta \tilde{\psi} - \beta \nabla^2 \delta \tilde{\psi}) d\Omega - \int_{\partial\Omega} \frac{\partial \beta}{\partial n} \delta \tilde{\psi} ds \\ &= - \int_{\Omega} Re(\nabla \omega \times \nabla \gamma) \delta \tilde{\psi} \mathbf{e}_z d\Omega - \int_{\Omega} \beta \delta \omega d\Omega \\ &= - \int_{\Omega} Re(\nabla \times (\gamma \nabla \omega)) \delta \tilde{\psi} \mathbf{e}_z d\Omega - \int_{\Omega} \beta \delta \omega d\Omega \\ &= - \int_{\Omega} Re \gamma \nabla \omega \cdot \nabla \times \delta \tilde{\psi} \mathbf{e}_z - \int_{\partial\Omega} Re \mathbf{n} \times (\gamma \nabla \omega) \delta \tilde{\psi} \mathbf{e}_z - \int_{\Omega} \beta \delta \omega d\Omega \\ &= \int_{\Omega} Re \gamma \nabla \omega \cdot \delta u d\Omega - \int_{\Omega} \beta \delta \omega d\Omega\end{aligned}$$

In addition, using the sensitivity equation of ω , we have

$$\int_{\Omega} \gamma \nabla^2 \delta \omega \, d\Omega - \int_{\Omega} \operatorname{Re} \gamma \nabla \cdot (u \delta \omega + \omega \delta u) \, d\Omega = 0$$

and, by the no-penetration condition, we have

$$\begin{aligned} \int_{\Omega} \operatorname{Re} \gamma \nabla \cdot (u \delta \omega + \omega \delta u) \, d\Omega &= - \int_{\Omega} \operatorname{Re} (u \delta \omega + \omega \delta u) \cdot \nabla \gamma \, d\Omega \\ &= - \int_{\Omega} \operatorname{Re} \delta \omega (u \cdot \nabla) \gamma \, d\Omega - \int_{\Omega} \operatorname{Re} \omega (\delta u \cdot \nabla) \gamma \, d\Omega \end{aligned}$$

Applying the second Green's identity, it renders us

$$\begin{aligned} \int_{\Omega} \gamma \nabla^2 \delta \omega \, d\Omega &= \int_{\Omega} \delta \omega \nabla^2 \gamma \, d\Omega + \int_{\partial\Omega} \gamma \frac{\partial \delta \omega}{\partial n} \, ds - \int_{\partial\Omega} \delta \omega \frac{\partial \gamma}{\partial n} \, ds \\ &= \int_{\Omega} (\beta - \operatorname{Re} (u \cdot \nabla) \gamma) \delta \omega \, d\Omega - \int_{\partial\Omega} \frac{\partial \gamma}{\partial n} \delta \omega \, ds \end{aligned}$$

Thus

$$\begin{aligned} \int_{\Omega} \gamma \nabla^2 \delta \omega \, d\Omega - \int_{\Omega} \operatorname{Re} \gamma \nabla \cdot (u \delta \omega + \omega \delta u) \, d\Omega &= \int_{\Omega} \beta \delta \omega \, d\Omega - \int_{\partial\Omega} \frac{\partial \gamma}{\partial n} \delta \omega \, ds + \int_{\Omega} \operatorname{Re} \omega (\delta u \cdot \nabla) \gamma \, d\Omega \\ &= 0 \end{aligned}$$

Meanwhile, we have

$$\int_{\Omega} \gamma \nabla \omega \cdot \delta u \, d\Omega = \int_{\Omega} \nabla \cdot (\gamma \omega \delta u) \, d\Omega - \int_{\Omega} \omega \nabla \gamma \cdot \delta u \, d\Omega - \int_{\Omega} \gamma \omega \nabla \cdot \delta u \, d\Omega$$

Since the flow is incompressible, thus

$$\nabla \cdot (u + \varepsilon \delta u) = 0$$

which implies

$$\nabla \cdot \delta u = 0$$

Therefore

$$\begin{aligned} \int_{\Omega} \gamma \nabla \omega \cdot \delta u \, d\Omega &= \int_{\Omega} \nabla \cdot (\gamma \omega \delta u) \, d\Omega - \int_{\Omega} \omega \nabla \gamma \cdot \delta u \, d\Omega \\ &= - \int_{\Omega} \omega (\delta u \cdot \nabla) \gamma \, d\Omega \end{aligned}$$

and we obtain

$$- \operatorname{Re} \int_{\Omega} \omega (\delta u \cdot \nabla) \gamma \, d\Omega = \int_{\Omega} \beta \delta \omega \, d\Omega - \int_{\partial\Omega} \frac{\partial \gamma}{\partial n} \delta \omega \, ds$$

Substitute this expression into the derivative of first variation of the objective functional, we have

$$\begin{aligned}\left.\frac{dJ}{d\varepsilon}\right|_{\varepsilon=0} &= \int_{\Omega} \beta \delta\omega \, d\Omega - \int_{\partial\Omega} \frac{\partial\gamma}{\partial n} \delta\omega \, ds - \int_{\Omega} \beta \delta\omega \, d\Omega + \int_{\partial\Omega} \alpha\eta \delta\eta \, ds \\ &= \int_{\partial\Omega} \left(-\frac{\partial\gamma}{\partial n} + \alpha\eta\right) \delta\eta \, ds \\ &= \nabla_{\eta} J \cdot \delta\eta\end{aligned}$$

which is the gradient of the objective functional.

Thought

- For the finite dimensional case, the adjoint operator is the matrix conjugate transpose, but our case is infinite dimensional.
- Under what situation, the adjoint system may have multiple solutions? What will happen if the solution is not unique?

Reference:

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